Date: 24/10/24.

Q VLSM Subnetting:
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Q) Question #1:

Network: 200.10.10.0/24
Sales: 40 hosts
HR: 12 hosts
WAN-link: 2 hosts.

part(i):

Network	Hosts	Hosts+2	Block	prefix	Mask.
Sales	40	42	64	32-6=/26	255.255.255.192
HR	12	14	16	32-4=/28	255.255.255.240
WAN-link	2	4	4	32-2=/30	255.255.255.252

Network-ID: prev. Broadcast #.
Usable: Block-size-2.

part(ii)

Network	Network-ID	Usable IP-range	Broadcast
Sales	200.10.10.0	200.10.10.1-.62	200.10.10.63
HR	200.10.10.64	200.10.10.65-.78	200.10.10.79
WAN-Link	200.10.10.80	200.10.10.81-.82	200.10.10.83

part(iii)
Next N-ID: 83+1=84.
256-84=172 for addresses.

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Part (iv).

Total allocated Block size
64+16+4 = 84.
84 ≤ 256
Therefore, all three fit inside the /24 network with 172 free addresses

AM

3) Question #2:

Given network : 192.168.100.0/24.

Network	Host	Host+2	Block size	Prefix	Subnet Mask
Computer Lab	60	62	64	24+2=26	192.168.100.192
Library	25	27	32	24+5=/27	192.168.100.224
office	10	12	16	24+4=/28	192.168.100.240
WAN link	2	2	4	24+1=/30	192.168.100.252

Part (ii) :-

Network	Network ID	Usable network IP range	Broadcast
Computer Lab	192.168.100.0	192.168.100.1 – 192.168.100.62	192.168.100.63
Library	192.168.100.64	192.168.100.65 – 192.168.100.95	192.168.100.95
office	192.168.100.96	192.168.100.97 – 192.168.100.110	192.168.100.111
WAN link	192.168.100.112	192.168.100.113 – 192.168.100.114	192.168.100.115

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Next-Network-ID : 115+1 = 116.

256 - 116 = 140 free address

c) Total block size

64+32+16+4 = 116.

116 < 256

2) Yes. All networks fits inside the /24 network.

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